

CSC310 Data Structures and Algorithm Analysis (Dr. Ng)

Programming Assignment: Building Binary Tree

Write a program to build a binary tree (using pointers) from a string of characters and print out the binary tree in preorder, inorder, postorder and level-order.

Part I:

Using pointer implementation of Binary Tree (BinaryTree), write a function:

```
BinaryTree BuildTree( string s);  
// returns the root of the binary tree represented by the String s.
```

The input string is a **preorder** order representation of a binary tree. In particular, a dot/period signifies an empty subtree in that branch. There **should not** be any node that holds the character ".". Character "." is not part of the allowable character in the node. You have to check whether the input string is a correct representation of a possible binary tree or not. You are **NOT** building a binary search tree. The binary tree that you build is not necessarily full or complete.

You can assume that each node consists of **one character** only. However, **your program should account for user input errors, especially whether the input string is a possible representation of a binary tree.**

Part II:

Using the binary tree T built in Part I, write and test the following functions.

```
bool IsBST( BinaryTree T)  
// determine whether the binary tree T is a binary search tree or not.  
// You cannot call an inorder function or use any sorting routine to assist you.
```

```
int node_count( BinaryTree T)  
// returns the number of nodes in the binary tree T.
```

```
int height( BinaryTree T)  
// returns the height of the binary tree T.
```

```
Int IsFullBT( BinaryTree T)  
// returns TRUE if the binary tree T is a full binary tree, false otherwise.
```

```
Int IsCompleteBT( BinaryTree T)  
// returns TRUE if the binary tree T is a complete binary tree, false otherwise.
```

You should use C++ for this assignment. You cannot call any library functions except those that deal with I/O and string handling functions.

Your program should execute interactively on your computer. Please test your program thoroughly and make sure it is free of errors. Points will be deducted for each error that I discovered for you.

Please also pay attention to program design and follow all the good programming practice. I reserve the right to deduct points based on these criteria.

Submit: Program design
Hierarchical diagrams/ Structure charts
Procedure specification
Program listing
Testing cases with Justification
Results of test cases

Bonus: (5 points for a, 10 points for b, 15 points for c and 20 points for d)

(**no machine generated codes or codes from internet**)

Without using extra data structure, graphically print the tree built from pointer implementation,

- a. horizontally without arcs connecting the nodes
- b. horizontally with arcs connecting the nodes
- c. vertically without arcs connecting the nodes
- d. vertically with arcs connecting the nodes

Bonus points will count only if the basic assignment has been completed and correct at an acceptable level.

Optional 1:

Repeat Part I but build a binary tree using postorder input.

Optional 2:

Extend Part I such that the node does not necessarily consist of a single character, but rather a string of characters separated by "|".

For example:

Build a binary tree with the following preorder:

= | animal | . | . | + | dog | . | . | cat | . | .

Optional 3:

Given a string which is the inorder representation of a binary tree, build **ALL** possible binary trees with the given inorder traversal.